# Lab 7: Seasonal ARIMA

MATH840 | Time Series | Week 8 — graded assignment 6 of 7

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**This week: accuracy is marked again**

Lab 6 measured your ARIMA without marking it, because a model without seasonal terms could not be expected to beat every benchmark. This week the seasonal terms are in, every series in the pool is one a seasonal ARIMA is known to beat, and the accuracy points are back: **3 of the 8**.

**Open the notebook**

The template: Open in Colab

Click **Copy to Drive** before you start working, otherwise nothing you write is saved.

The practice session's four cases, with the same helpers: Open in Colab

## The task

Forecast **your series from Week 4** with seasonal ARIMA, $\mathrm{ARIMA}(p, d, q)(P, D, Q)_m$ — the *ARIMA Models, Part 2* lecture. The order matters, and it is marked:

1. **Difference** — $D$ and $d$ from the tests, $\log y$ if the seasonal swing grows with the level.
2. **Read** the ACF and PACF of the differenced series: $p, q$ at lags $1, 2, 3$; $P, Q$ at lags $m, 2m, 3m$.
3. **Specify by hand** at least two candidates — then add `AutoARIMA`.
4. **Decide** on a rolling origin, check the residuals with $p + q + P + Q$, submit.

ARIMA only: no ETS, no regression, no averaging with other methods. Combining is next week's lecture.

**This one counts**

8 points. **Deadline: 23:59 today**, on Moodle. The accuracy score is published the next morning; marks and comments come back before the next practice session.

## What you are marked on

| Section | Criteria | Points |
|---|---|---|
| 1 | EDA and visualisation | 1 |
| 2 | Forecast accuracy against your benchmark | 3 |

| Section | Criteria | Points |
|---|---|---|
| 3 | Justification block | 3 |
| 4 | Code quality and reproducibility | 1 |
| | **Total** | **8** |

The notebook has one section per row. **Keep those headings.**

## 📊 Section 1 — EDA and visualisation (1 point)

- A time plot, read for trend, season, and whether the seasonal swing grows with the level — your case for or against `log=True`.
- The validation split on a plot: history, validation window, your chosen model's forecast and 80% interval over it.
- One sentence per plot. **A plot with no reading attached earns nothing.**

## 🎯 Section 2 — Forecast accuracy (3 points)

Scored from your submission file on the `h` observations after your history that nobody has seen:

$$\text{skill} = \frac{\text{MASE of your forecast}}{\text{MASE of your benchmark}}$$

| Skill | Points |
|---|---|
| below 1 — you beat your benchmark | 3 |
| 1 to 1.1 — within 10% | 2 |
| above 1.1 | 1, and 3.3 must say why |

> ⓘ **What the pool guarantees, and what it does not**
>
> Every series in the pool was issued only if `AutoARIMA` with `season_length=m`, **fitted on the whole raw history**, beats its benchmark on the hidden horizon. That is a guarantee about that one model: not about every seasonal ARIMA, not about `AutoARIMA` on $\log y$, and not about the orders `AutoARIMA` found on `train`.
>
> Measured on the whole pool with this template: choosing among the airline model and `AutoARIMA`, each with and without a log, by `rolling_mase`, beats the benchmark on **89%** of series, comes within 10% on another **9%**, and loses on **2%**. `AutoARIMA` on the raw history alone beats it on all 140 — the screen guarantees that.
>
> Hand-specified models and logs are what the rubric asks you to try — follow your evidence, and say in 3.2 what it was.

## ✍ Section 3 — Justification block (3 points)

The part a human reads closely. Roughly a paragraph each, with numbers from your own cells.

| Item | Points | What earns them |
|---|---|---|
| 3.1 | 1 | **The specification, before the machine.** $D$, $d$ and the log decision, each tied to a test or a plot. The lags at the start of the ACF/PACF that named $p$, $q$, and the seasonal lags that named $P$, $Q$. |
| 3.2 | 1 | **What `AutoARIMA` chose, and what you submitted.** Where it agrees or disagrees with your reading, and why. Your candidates with AICc and rolling MASE. The model you submitted and the evidence for it — and if it is not `AutoARIMA`'s choice, what made you leave it. |
| 3.3 | 1 | **What is left, and what you expect.** Ljung-Box with $p + q + P + Q$ degrees of freedom, read. The residual autocorrelation at lag $m$, against your Lab 6 model. The interval's coverage on the validation window. Your expected skill — and if you expect to lose, why. |

> **Important**
>
> Marked on the reasoning. A hand reading that `AutoARIMA` overrules, with a clear account of why, scores full marks in 3.1 and 3.2. So does a model that loses to the benchmark, if 3.3 explains the loss with numbers.

> **⚠ Where this goes wrong**
>
> - **Reading only the first lags.** $P$ and $Q$ live at $m, 2m, 3m$ — the shaded lags on the template's plot. A spike at lag 12 with nothing at 24 is a seasonal MA(1); a decay over 12, 24, 36 is seasonal AR.
> - **Two seasonal differences.** $D \leq 1$, always, and $d + D \leq 2$.
> - **Comparing AICc across different $d$, $D$ or log choices.** Across those, `rolling_mase` decides.
> - **Ljung-Box with $p + q$ only.** The seasonal terms spend degrees of freedom too: $p + q + P + Q$.
> - **Freezing `AutoARIMA`'s orders from `train`.** If you choose `AutoARIMA`, submit it refitted on all of `y` — that is the model the pool guarantee is about.
> - **Forecasting from `train`.** The submission is refitted on the whole history.
>
> 🕮 FPP3 §9.9 seasonal ARIMA · §9.10 ARIMA vs ETS · FPP the Pythonic Way, ch. 9

## 🧹 Section 4 — Code quality and reproducibility (1 point)

- The notebook runs top to bottom on a fresh kernel without errors.
- The submission file is written **by the notebook**, not edited by hand.
- No leakage: candidates are compared on `train` / `valid` and `rolling_mase`; only the final model is fitted on all of `y`.
- Every number in your text comes from a cell.

## Submission

Upload three files to the Week 8 activity on Moodle **before 23:59 today**:

- `SURNAME_lab08_forecast.csv` — written by the last cell of the template. Exactly $h$ rows and these columns:

| Column | What |
|---|---|
| `student_id` | your identifier |
| `code` | your series code, the same as before |
| `ds` | the dates you are forecasting, in order |
| `yhat` | your point forecast |
| `yhat_lo_80`, `yhat_hi_80` | your model's 80% interval |

- `SURNAME_lab08.pdf` — the report, with the justification block;
- `SURNAME_lab08.ipynb` — the source.

> ⚠ **A file that cannot be read scores 0 for accuracy**
>
> Exactly $h$ rows, the dates asked for, no `NaN`, the interval around the point forecast. The template's check cell is the same one the scorer uses — if it passes, your file is fine.

## How long this should take

Ninety minutes if you follow the practice session's first case. A seasonal `AutoARIMA` inside `rolling_mase` refits three times; on a monthly series a full table takes a few minutes, so run it once and keep the result.

## For next week

**Lab 8** is the last weekly lab: the *Advanced Forecasting Methods* lecture, already published — combining the models you have built in Labs 5–7. Keep your Lab 5 and Lab 7 notebooks: next week you average their forecasts.